Setup v18:

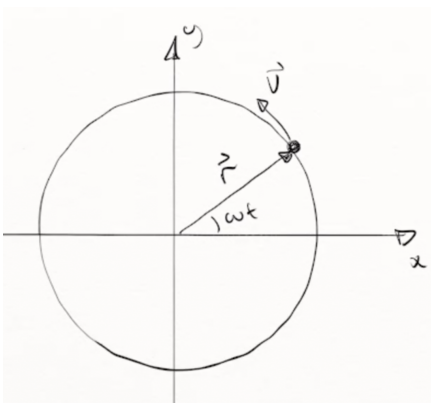
Review

Special Relativity is much like normal relative motion, just with a weird demand that some speeds must be the same regardless of reference frame, so it pays to refresh your memory of relative motion and of coordinate systems.

Remember, a mass of equations is not an answer. A reasoned argument is an answer. Your job is to learn to think these things through; show me how you think so I can critique the skill you're trying to hone.

1) Uniform circular motion

Given a particle moving in uniform circular motion with the given $\vec{r}[t]$, determine the velocity and acceleration, $\vec{v}[t]$ and $\vec{a}[t]$, as well as their magnitudes, v and a. Express a three ways: without ω, without v, and without r.



Start with

$$\vec{r}[t] = \{r_0 \operatorname{Cos}[\omega\, t],\ r_0 \operatorname{Sin}[\omega\, t]\} \tag{1}$$

What are the directions of the velocity and acceleration? If you know the acceleration as a function of all the other things, what happens to the acceleration if you double r? If you double ω?

2) Plane, air, ground

A large passenger airplane is attempting to land on a mile-long runway that is oriented exactly south to north. At high altitude the wind in the jet stream is blowing at a v_{stream} from the west, but at the ground level there is a strong wind blowing from θ_g east of north at v_g. The plane's airspeed (the speed relative to the air immediately in front of the plane) the instant before it lands is v_p. At what angle, relative to north, does the plane have to orient its nose so that it can approach the runway along a direct south to north trajectory? What conclusions can you draw from the answer?

3) Coordinate systems and kinematic relations

Consider the object thrown off a cliff and over a wall situation in the figure. The constant acceleration 2D equations of motion is always given as

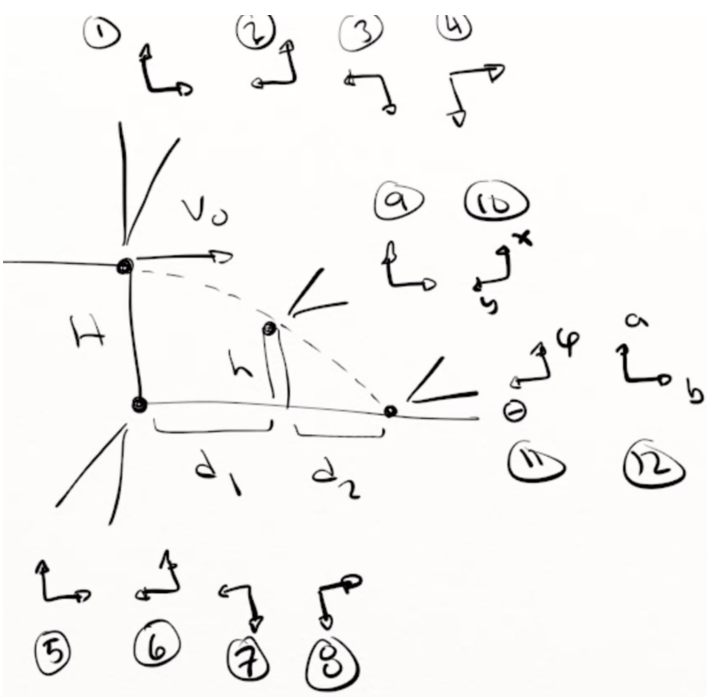
$$x[t] = x_0 + v_{x0}\, t \qquad y[t] = y_0 + v_{y0}\, t + \frac{1}{2}\, a\, t^2 \tag{2}$$

But this is the generic result, for different coordinate systems, it must be changed. Rewrite the equation for each of the coordinate systems in the image. Be sure to take into account the origin, axis directions, and coordinate variables. Let the magnitude of acceleration be g. Note that at $t = 0$, the object should be at $\{x_0, y_0\}$.

If no axis variables are given use x and y for horizontal and vertical. So, for #10, they're swapped.

For 1 through 4, the origin is at the top of the hill.

You know all the things, H, h, d_1, d_2, and v_0.

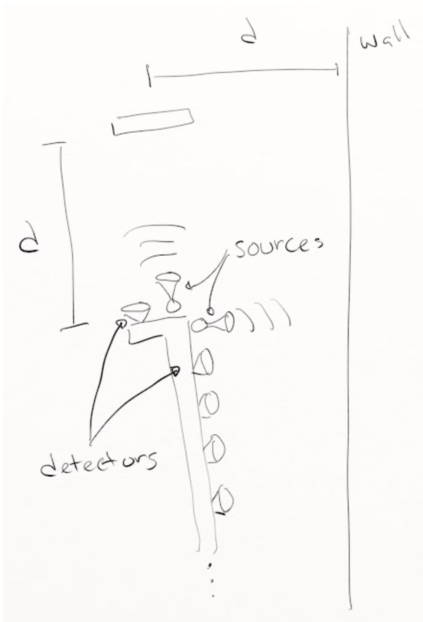


Follow the arguments

4) The Michelson-Morley experiment in air

The Michelson-Morley experiment attempted to measure the velocity of the Earth through the luminiferous ether, the postulated medium which waved when light waves moved from one point to another. It conclusively showed that there was no such motion, no matter where on Earth (or off Earth, once we were able to put satellites in space) one was. Since the Earth is spinning, this posed a rather serious conundrum.

We here work through the argument of what was supposed to happen. We ignore issues related to light by conceiving of a version of the experiment based on sound waves.



We have an apparatus consisting of a pair of clickers and a set of microphones along with a barrier, d away, hooked to them. This is a distance d away from a wall.

(a) Consider first nothing moving. A click is emitted from the two clickers. Sound travels to the wall, reflects off and returns. Sound travels up to the barrier, reflects off and returns. Given the speed of sound, v_s, how long does it take to make the round trips to the wall and barrier, Δt_w and Δt_b?

(b) Now let a wind be blowing downward. One might be tempted to say that the click moving to the wall must travel a longer distance, but this is not true. The sound, once launched into the air moves *in the air* at v_s toward the wall. So the round trip time is the same (we have microphones spread out so the sound doesn't need to travel "up wind"). But what is the time up to the barrier, t_{up}? And the time down again, t_{down}? So the round trip time is?

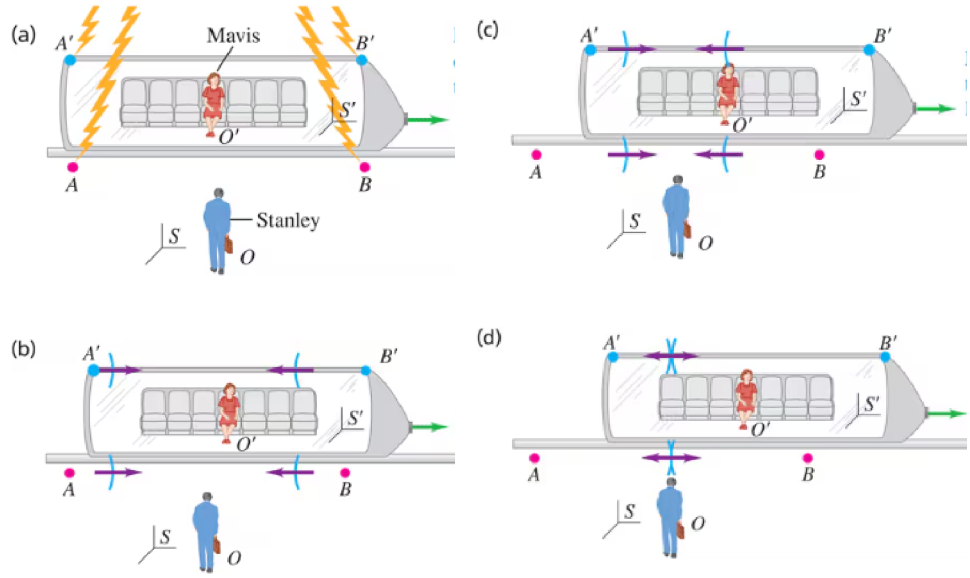
(c) It is this difference in time, which you should have found to depend on the speed of the wind, that Michelson and Morley were looking for, just with light. What if the wind is blowing upward, what is the round trip time then?

5) Train and lightning

Two lightning bolts strike a passenger car, one near each end. Each bolt leaves a mark on the car and one on the ground at the instant the bolt hits. The points on the ground are labeled A and B, and the corresponding points on the car are A' and B' Stanley is stationary on the ground at O midway between A and B, Mavis is moving with the train at O' in the middle of the passenger car, midway between A' and B'. Both Stanley and Mavis see both light flashes emitted from the points where the lightning strikes.

Here are four pictures of the process. Argue that Stanley and Mavis disagree about whether the lightning strikes were simultaneous by discussing what each picture demonstrates. Note that there is a fact that Mavis, who knows about special relativity, makes use of that is not represented in the figures.

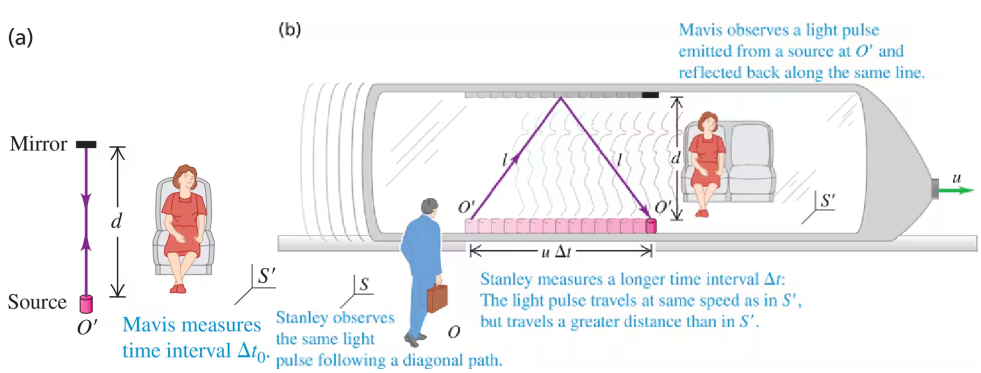
Out[∘]=



6) Time dilation

Let us rework the time dilation argument. Mavis, still in the train car speeding past Stanley, has a light source mounted on the floor and a mirror in the ceiling. A light pulse is sent upward and its return is recorded, giving the time of flight up and back down. Stanley, though, must see something different, as in the figure.

Out[∘]=



(a) Deduce the round trip *distance* for the light in both Mavis' and Stanley's frame.

(b) What is the round trip *time* for Stanley? Mavis?

(c) Both relations should have a height, d, in them. Clearly, it would be good if the relationship between the two times were expressed without needing to know the distance (in whatever frame). Eliminate d.

(d) Can you, instead, eliminate the speed from both times?

(e) Solve for the round trip time as experienced by Stanley and argue that, no matter what the speed is, the time must be longer than it is for Mavis.

Problems

These are from the Sears and Zemansky text. They are not in order of the material we covered so, for instance, if you're looking at this right after it was posted, you may not know what proper time and proper length are in the first question.

7) Propriety in time and space intervals

You are standing on a train platform watching a high-speed train pass by. A light inside one of the train cars is turned on and then a little later it is turned off. (a) Who can measure the proper time interval for the duration of the light: you or a passenger on the train? (b) Who can measure the proper length of the train car: you or a passenger on the train? (c) Who can measure the proper length of a sign attached to a post on the train platform: you or a passenger on the train?

8) The limits of *non*-simultaneity

Two events occur at the same space point in a particular inertial frame of reference and are simultaneous in that frame. Is it possible that they may not be simultaneous in a different inertial frame?

9) When shorter, when longer?

A high-speed train passes a train platform. Larry is a passenger on the train, Gene is standing on the train platform, and David is riding a bicycle toward the platform in the same direction as the train is traveling. Compare the length of a train car as measured by Larry, Gene, and David.

10) Events and the events of their observation

Consider a variation of the train-and-lightning example where the two lightning bolts are simultaneous *to an observer on the train*. Show that they are *not* simultaneous to an observer on the ground. Which lightning strike does the ground observer measure to come first?

Hint: My answer is one sentence.

11) The reality of reference frames

Muons are unstable subatomic particles that decay to electrons with a mean lifetime of 2.2 μs. They are produced when cosmic rays bombard the upper atmosphere about 10 km above the earth's surface, and they travel very close to the speed of light. The problem we want to address is why we see any of them at the earth's surface. (a) What is the greatest distance a muon could travel during its 2.2 μs lifetime? (b) According to your answer in part (a), it would seem that muons could never make it to the ground. But the 2.2 μs lifetime is measured in the frame of the muon, and muons are moving very fast. At a speed of 0.999 c what is the mean lifetime of a muon as measured by an observer at rest on the earth? How far would the muon travel in this time? Does this result explain why we find muons in cosmic rays? (c) From the point of view of the muon, it still lives for only 2.2 μs so how does it make it to the ground? What is the thickness of the 10 km of atmosphere through which the muon must travel, as measured by the muon? Is it now clear how the muon is able to reach the ground?

12) Same situation, because there really is only one

The negative pion, π^-, is an unstable particle with an average lifetime of 26 ns (measured in the rest frame of the pion). (a) If the pion is made to travel at very high speed relative to a laboratory, its average lifetime

is measured in the laboratory to be 400 ns. Calculate the speed of the pion expressed as a fraction of c. (b) What distance, measured in the laboratory, does the pion travel during its average lifetime?

13) Which time is dilated?

The average life span in the United States is about 70 years. Does this mean that it is impossible for an average person to travel a distance greater than 70 light-years away from the earth?